Spacescape

Overview:

Spacescape is a vertical platformer game about an astronaut stuck on a planet who is trying to escape and reach its mothership. The astronaut works to reach the ship by jumping across platforms made from abandoned debris or asteroid fragments while also dodging or destroying varying sized pieces of asteroids falling from above. The player controls the movement of the astronaut and jumps between platforms. The game becomes more challenging as the astronaut gets higher. The asteroids will fall faster and become more frequent. The player can also temporarily become invincible to the falling asteroids by activating a power-up, which occasionally spawns on platforms. Additionally, the platforms will become harder to reach by the player. The game is designed to be a continuous climb so players can increase their score. The score is based on the astronaut's height relative to the starting planet and the number of asteroids destroyed. If creating an infinite level becomes too difficult, the game will instead use a target score.

Controls: The controls will be mouse and keyboard.

- **Movement:** The 'A' and 'D' keys will move the astronaut to the left and right.
- **Jump:** The space bar will allow the astronaut to jump onto the higher platforms.
- **Aim:** The mouse will control the direction of the astronaut's weapon.
- **Fire:** The left click will fire the projectile. The astronaut will always be able to shoot, but there will be a cooldown between shots. If aiming in multiple directions becomes too difficult, the weapon will only fire straight upward.
- **Shield**: When the player picks up the power-up, the 'E' key will activate a shield that will temporarily make the astronaut invincible from falling asteroids.

Art Assets Needed:

- **Astronaut**: An astronaut with walking, jumping, and death animations to be the player.
- **Weapon:** Gun with projectiles that are shown when the player shoots.
- **Debris**: Pieces of wreckage to serve as the platforms the astronaut jumps on.
- **Asteroid:** Asteroids of varying sizes to fall from the top of the screen.
- **Background:** Scrolling space background with stars and planets.
- **HUD:** Indicators showing shield power-up status and score.
- **Screens**: Title and game over screens.

Audio Assets Needed:

- **Astronaut:** Jumping, landing, falling, and death.
- **Weapon:** Projectile firing, projectile impact.
- **Asteroid:** Impact (large, small).
- **Shield:** Pickup, activation, impact, and expiration
- **Game:** Game over (when astronaut dies).

Game Flow:

The player loads into the game from the title screen and the player begins on the bottom level, a planet. The astronaut climbs through space, jumping through the various platforms, while also dodging falling asteroids. The player uses the mouse to aim and left click to shoot the falling asteroids. The player can collect a shield power-up and activate it with ‘E’, which will grant invincibility to asteroids temporarily. As the astronaut climbs higher, the game becomes more difficult with faster and more frequent asteroids and harder to reach platforms. The score increases throughout the game by increasing height and destroying falling debris. Falling or being hit by falling asteroids will end the game and the game will restart.

Concerns:

- **Weapon aiming and animation:** It may be challenging trying to animate the astronaut and weapon correctly for the possible shooting and aiming direction around the astronaut. Some animations may make it look awkward if the astronaut is able to shoot across and behind its body. If this becomes too difficult, I may limit the weapon to only being able to shoot upward instead of allowing it to shoot in the direction of the mouse.
- **Moving platforms:** Moving platforms may be difficult to implement because the astronaut must be able to land on them without sliding off or passing through them. The astronaut should move with the platform when it lands on it. Additionally, I would need to test if every jump is doable from these platforms. If manual jumping makes the game too easy, I may try automatic jumping like other classic games to make timing and placement more challenging.
- **Camera Scrolling and Boundaries:** Since the astronaut should be jumping upward, the camera must follow the player while keeping the next available platforms visible. I need to make sure that platforms off the screen and already passed by the player are removed to ensure there are not too many assets, which would harm performance, and so that they cannot be landed on. It may also be challenging having the astronaut wrap around the screen to the opposite side once it hits the edge of the screen.
- **Difficulty Progression:** It may be difficult to try and make the game infinite because making safe and reachable platforms could increase the project’s scope. If this becomes too complicated, I will try and use my second idea of having a target score as the main objective instead. It may have a timer so players could compete to reach it in the fastest time. It also may be challenging to balance the difficulty and speed so that the game is still challenging without preventing players from being able to react in time.